CODE CHALLENGE: Solve the Profile HMM Problem.

Input: A threshold *θ*, followed by an alphabet *Σ*, followed by a multiple alignment *Alignment* whose strings are formed from *Σ*.

Output: The transition matrix followed by the emission matrix of HMM(*Alignment, θ*).

Note: Your matrices can be either space-separated or tab-separated.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/hmm/ProfileHMM.txt)

Sample Input:

```
0.289
--------
A B C D E
--------
EBA
E-D
EB-
EED
EBD
EBE
E-D
E-D
```

Sample Output:

```
	S	I0	M1	D1	I1	M2	D2	I2	E
S	0	0	1.0	0	0	0	0	0	0
I0	0	0	0	0	0	0	0	0	0
M1	0	0	0	0	0.625	0.375	0	0	0
D1	0	0	0	0	0	0	0	0	0
I1	0	0	0	0	0	0.8	0.2	0	0
M2	0	0	0	0	0	0	0	0	1.0
D2	0	0	0	0	0	0	0	0	1.0
I2	0	0	0	0	0	0	0	0	0
E	0	0	0	0	0	0	0	0	0
--------
	A	B	C	D	E
S	0	0	0	0	0
I0	0	0	0	0	0
M1	0	0	0	0	1.0
D1	0	0	0	0	0
I1	0	0.8	0	0	0.2
M2	0.143	0	0	0.714	0.143
D2	0	0	0	0	0
I2	0	0	0	0	0
E	0	0	0	0	0
```

Time Limit: `5 mins`

To solve this problem please visit stepic.org/lesson/-11632/step/2

We saw in a previous chapter that multiplication by zero can cause problems when finding a most likely object. Our workaround there was to introduce pseudocounts to prevent this multiplication.

In the case of a profile HMM, for example, say that we transition from state *A* to state *B* with probability 0.833, state *A* to state *C* with probability 0.167, and state *A* to state *D* with probability 0. In the current setup, it is not possible for a path passing from state *A* to state *D* to have nonzero weight. Yet adding a pseudocount of 0.01 to each of these probabilities, then renormalizing, results in the following updated transition probabilities after rounding to three decimal places:

Pr(*A*->*B*) = (0.833 + 0.01) /((0.833 + 0.01) + (0.167 + 0.01) + (0 + 0.01)) = 0.818
Pr(*A*->*C*) = (0.167 + 0.01) /((0.833 + 0.01) + (0.167 + 0.01) + (0 + 0.01)) = 0.172
Pr(*A*->*D*) = (0 + 0.01) / ((0.833 + 0.01) + (0.167 + 0.01) + (0 + 0.01))= 0.010

This discussion brings us to the following problem.

Profile HMM with Pseudocounts Problem: *Construct a profile HMM with pseudocounts from a multiple alignment.*
Input: A multiple alignment *Alignment*, a threshold value θ, and a pseudocount value σ.
Output: HMM(*Alignment*, θ), in the form of transition and emission matrices.

CODE CHALLENGE: Solve the Profile HMM with Pseudocounts Problem.

Input: A threshold θ and a pseudocount σ, followed by an alphabet Σ, followed by a multiple alignment *Alignment* whose strings are formed from Σ.

Output: The transition and emission matrices of HMM(*Alignment*, θ, σ).

Note: Your matrices can be either space-separated or tab-separated.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/hmm/ProfileHMMPseudocounts.txt)

Sample Input:

```
0.358 0.01
--------
A B C D E
--------
A-A
ADA
ACA
A-C
-EA
D-A
```

Sample Output:

```
        S       I0      M1      D1      I1      M2      D2      I2      E
S       0       0.01    0.819   0.172   0       0       0       0       0
I0      0       0.333   0.333   0.333   0       0       0       0       0
M1      0       0       0       0       0.398   0.592   0.01    0       0
D1      0       0       0       0       0.981   0.01    0.01    0       0
I1      0       0       0       0       0.01    0.981   0.01    0       0
M2      0       0       0       0       0       0       0       0.01    0.99
D2      0       0       0       0       0       0       0       0.5     0.5
I2      0       0       0       0       0       0       0       0.5     0.5
E       0       0       0       0       0       0       0       0       0
--------
        A       B       C       D       E
S       0       0       0       0       0
I0      0.2     0.2     0.2     0.2     0.2
M1      0.771   0.01    0.01    0.2     0.01
D1      0       0       0       0       0
I1      0.01    0.01    0.327   0.327   0.327
M2      0.803   0.01    0.168   0.01    0.01
D2      0       0       0       0       0
I2      0.2     0.2     0.2     0.2     0.2
E       0       0       0       0       0
```

Time Limit: `5 mins`

To solve this problem please visit stepic.org/lesson/-11632/step/4

Sequence Alignment with Profile HMM Problem: *Align a new sequence to a family of sequences using a profile HMM.*

Input: A multiple alignment *Alignment*, a threshold value θ, a pseudocount value σ, and a string x.

Output: An optimal hidden path emitting x in HMM(*Alignment*, θ, σ).

CODE CHALLENGE: Solve the Sequence Alignment with Profile HMM Problem.

Input: A string *x* followed by a threshold *θ* and a pseudocount *σ*, followed by an alphabet *Σ*, followed by a multiple alignment *Alignment* whose strings are formed from *Σ*.

Output: An optimal hidden path emitting *x* in HMM(*Alignment, θ, σ*).

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/hmm/AlignmentWithHMMProfile.txt)

Sample Input:

```
AEFDFDC
--------
0.4 0.01
--------
A B C D E F
--------
ACDEFACADF
AFDA---CCF
A--EFD-FDC
ACAEF--A-C
ADDEFAAADF
```

Sample Output:

```
M1 D2 D3 M4 M5 I5 M6 M7 M8
```

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11632/step/6

HMM Parameter Estimation Problem: *Find optimal parameters explaining the emitted string and hidden path of an HMM.*

Input: A string $x = x_1 \ldots x_n$ emitted by a k-state HMM with unknown transition and emission probabilities following a known hidden path $\pi = \pi_1 \ldots - \pi_n$.

Output: A transition matrix *Transition* and an emission matrix *Emission* that maximize $Pr(x, \pi)$ over all possible transition and emission matrices.

If we know both x and π, then we can compute empirical estimates for the transition and emission probabilities using a method similar to one we used for estimating parameters for profile HMMs. If $T_{l,k}$ denotes the number of transitions from state l to state k in the hidden path π, then we can estimate the probability $transition_{l,k}$ by computing the ratio of $T_{l,k}$ to the total number of transitions leaving state l,

$$transition_{l,k} = \frac{T_{l,k}}{\sum_{\text{all states } j} T_{l,j}}.$$

Likewise, if $E_k(b)$ denotes the number of times symbol b is emitted when the hidden path π is in state k, then we can estimate the probability $emission_k(b)$ as the ratio of $E_k(b)$ to the total number of emitted symbols from state k,

$$emission_k(b) = \frac{E_k(b)}{\sum_{\text{all symbols } c \text{ in the alphabet}} E_k(c)}.$$

It turns out that the above two formulas for computing $Transition$ and $Emission$ result in parameters solving the HMM Parameter Estimation Problem.

CODE CHALLENGE: Solve the HMM Parameter Estimation Problem.

Input: A string *x* of symbols emitted from an HMM, followed by the HMM's alphabet *Σ*, followed by a path *π*, followed by the collection of states of the HMM.

Output: A transition matrix *Transition* followed by an emission matrix *Emission* that maximize Pr(*x, π*) over all possible transition and emission matrices.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/hmm/HMMParameterEstimation.txt)

Sample Input:

```
yzzzyxzxxx
--------
x y z
--------
BBABABABAB
--------
A B C
```

Sample Output:

```
        A       B       C
A       0.0     1.0     0.0
B       0.8     0.2     0.0
C       0.333   0.333   0.333
--------
        x       y       z
A       0.25    0.25    0.5
B       0.5     0.167   0.333
C       0.333   0.333   0.333
```

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11632/step/8

HMM Parameter Learning Problem: *Estimate the parameters of an HMM explaining an emitted string.*

Input: A string $x = x_1 \ldots x_n$ emitted by an HMM with unknown transition and emission probabilities.

Output: A transition matrix *Transition* and an emission matrix *Emission* that maximize $\Pr(x, \pi)$ over all possible transition and emission matrices and over all hidden paths π.

Unfortunately, the HMM Parameter Learning Problem is intractable, and so we will instead develop a heuristic that is analogous to the Lloyd algorithm for *k*-means clustering. In that algorithm, we iterated two steps, "From Centers to Clusters",

$$(Data, ?, Centers) \rightarrow HiddenVector,$$

and "From Clusters to Centers",

$$(Data, HiddenVector, ?) \rightarrow Centers.$$

As for HMM parameter estimation, we begin with an initial random guess for *Parameters*. Then, we use the Viterbi algorithm to find the optimal hidden path π:

$$(x, ?, Parameters) \rightarrow \pi$$

Once we know π, we will question our original choice of *Parameters* and apply our solution to the HMM Parameter Estimation Problem to update Parameters based on x and π:

$$(x, \pi, ?) \rightarrow Parameters'$$

We then iterate over these two steps, hoping that the estimated parameters are getting closer and closer to the parameters solving the HMM Parameter Learning Problem:

$$(x, ?, Parameters) \rightarrow (x, \pi, Parameters) \rightarrow (x, \pi, ?) \rightarrow (x, \pi, Parameters') \rightarrow (x, ?, Parameters') \rightarrow (x, \pi', Parameters') \rightarrow (x, \pi', ?) \rightarrow (x, \pi', Parameters'') \rightarrow \ldots$$

This approach to learning the HMM's parameters is called **Viterbi learning**.

CODE CHALLENGE: Implement Viterbi learning for estimating the parameters of an HMM.

Input: A number of iterations *j*, followed by a string *x* of symbols emitted by an HMM, followed by the HMM's alphabet *Σ*, followed by the HMM's states, followed by initial transition and emission matrices for the HMM.

Output: Emission and transition matrices resulting from applying Viterbi learning for *j* iterations.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/hmm/ViterbiLearning.txt)

Sample Input:

```
100
--------
zyzxzxxxzz
--------
x y z
--------
A B
--------
        A       B
A       0.599   0.401
B       0.294   0.706
--------
        x       y       z
A       0.424   0.367   0.209
B       0.262   0.449   0.289
```

Sample Output:

```
        A       B
A       0.5     0.5
B       0.0     1.0
--------
        x       y       z
A       0.333   0.333   0.333
B       0.4     0.1     0.5
```

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11632/step/10

Soft Decoding Problem: *Find the probability that an HMM was in a particular state at a particular moment given its emitted string.*

Input: A string $x = x_1 \ldots x_n$ emitted by an HMM.

Output: The conditional probability $\Pr(\pi_i = k|x)$ that the HMM was in state k at step i given that it emitted x.

Recall that we computed $forward_{k,i}$ when solving the Outcome Likelihood Problem. If we reverse the edges of the Viterbi graph, we can compute

$$backward_{k,i} = \sum_{\text{all states } l} backward_{l,i+1} \cdot Weight_i(k,l).$$

Here we have that the reversed edge connecting (l, i+1) to (k, i) has weight $Weight_i(k,l) = transition_{k,l} \cdot emission_l(x_{i+1})$.

Combining the forward-backward algorithm described in the lecture with our solution to the Outcome Likelihood Problem for computing Pr(x) yields that

$$\Pr(\pi_i = k|x) = \frac{\Pr(\pi_i = k, x)}{\Pr(x)} = \frac{forward_{k,i} \cdot backward_{k,i}}{forward(sink)}$$

Profile HMM Problem: *Construct a profile HMM from a multiple alignment.*

Input: A multiple alignment *Alignment* and a threshold θ.

Output: HMM(*Alignment*, θ), in the form of transition and emission matrices.

CODE CHALLENGE: Solve the Soft Decoding Problem.

Input: A string x, followed by the alphabet Σ from which x was constructed, followed by the states *States*, transition matrix *Transition*, and emission matrix *Emission* of an HMM (Σ, *States*, *Transition*, *Emission*).

Output: An $|x|$ x $|States|$ matrix whose (i, k)-th element holds the conditional probability $\Pr(\pi_i = k|x)$.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/hmm/SoftDecoding.txt)

Sample Input:

```
zyxxxxyxzz
--------
x y z
--------
A B
--------
        A       B
A       0.911   0.089
B       0.228   0.772
--------
        x       y       z
A       0.356   0.191   0.453
B       0.040   0.467   0.493
```

Sample Output:

```
A       B
0.5438  0.4562
0.6492  0.3508
0.9647  0.0353
0.9936  0.0064
0.9957  0.0043
0.9891  0.0109
0.9154  0.0846
0.964   0.036
0.8737  0.1263
0.8167  0.1833
```

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11632/step/12